Design and testing of a low-cost electromyogram that uses a right leg driver circuit

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6 authors, including:

Ashley Stewart
University of Canterbury

Eva Hansenne

Design and Testing of a low-cost Electromyogram that Uses a Right Leg Driver Circuit

Benjamin C. Fortune, Ashley Stewart, Eva Hansenne, Lachlan McKenzie, Logan Chatfield, Chris G. Pretty
Department of Mechanical Engineering
University of Canterbury
Christchurch, New Zealand

***Abstract*—This paper presents the design and testing of a low cost ($30 unisolated, $75 isolated) electromyogram (EMG) circuit. The circuit is designed to utilise a Right Leg Driver (RLD) circuit: the RLD removes coupled electromagnetic interference from the subject. The benefit inherent with the use of the RLD is that it removes interference in real time, without attenuating the bio-potential signal. The circuit has been successfully tested to record bio-potential signals from multiple able-bodied subjects. A signal recorded from one subject's bicep brachii during an isometric contraction had a SNR of 57.8 dB**

***Index Terms*—exoskeleton, electromyography, circuit, design, right leg driver, RLD, rehabilitation**

I. Introduction

Electromyography (EMG) is the technique applied to measure the bio-potential signal produced by a contracting skeletal muscle. The bio-potential signal is the spatial and temporal summation of muscle fibre action potentials, know as the motor unit action potential (MUAP).

An EMG device was developed in the context of use alongside an exoskeleton with the intent of use for stroke rehabilitation research. The EMG device is designed to produce subject bio-feedback, which can be used for subject intention estimation, force estimation, limb angle estimation and determining the level of muscle activation [1], [2]; therefore, there is a need for low noise and low latency bio-potential sensing. These properties can be used to assist in exoskeleton control; however, these properties are widely used in other applications.

The use of EMG in assistive exoskeleton research is not uncommon [2]–[4]; however, when EMG has been used, the specific type of EMG device is typically not specified, not even to the degree of mentioning if the device is purposely constructed for their research, or if it is a commercially available device. Commercially available EMG devices which are capable of precision recordings of the bio-potential signal are expensive.

The recording of bio-potential signals is highly susceptible to noise, dominantly being coupled electromagnetic interference (EMI) from the surrounding AC mains power. With such a small signal amplitude ($\leq$10 mV pk-pk), which contains its useable energy within 0-500 Hz, but dominantly between 50-150 Hz, correct filtering is vital [5]. As the useable portions of the signal are those with larger amplitudes than the inherent noise, the attenuation of EMI (without hindering signal integrity) is one of the largest design considerations. This is commonly achieved through the use of a notch filter; however, this method attenuates desired frequencies. A design approach which is intended to solely attenuate the coupled interference is the use of a Right Leg Driver (RLD) circuit, outlined in section VII. EMG systems which utilise a RLD circuit have been designed before, thus the novelty behind this work is the implementation of a low cost EMG circuit design and schematic which utilises a RLD circuit.

This paper is divided as follows: section II gives a brief overview of the the origin of the MUAP; section III describes the advantages, limitations and possible methods to reduce the effects associated with these limitations; section IV is a brief review of current EMG designs and their limitations; section V is a high level design overview; section VI mentions the applied filtering methods and the associated reasons; section VII describes the fundamentals of the RLD circuit; section VIII explains why isolation is required for bio-detection devices; section IX provides circuit diagrams and circuit specifications; section X presents obtained data using the designed EMG device and how it compares to other results; section XI outlines the work which can still be done to further improve the current design; and section XII contains the conclusions.

II. The origin of the Bio-potential Signal

The motor unit and its associated alpha motor system are the basic level of the nervous system organisation of the muscle [6], [7]. This consists of the lower motor neuron, its axon and the various muscle fibres it innervates. The lower motor axon has multiple branches which allows it to connect to the motor end plate of the muscle fibre, creating neuromuscular synapses.

Voltage gated ion channels located in the axon control the flow of various ions within close proximity. The state of the voltage gated ion channels (either open or closed) are controlled by their electric potential. If there is a large enough spacial and temporal summation of excitatory potentials, the sodium ion voltage gated channels open, resulting in an inrush of sodium ions due to electrical and diffusion forces. The sudden influx of sodium ions depolarises the cell, causing an action potential. When an action potential travels down the axon and reaches the synapses, acetylcholine (ACh, a neurotransmitter) is released. The release of the ACh causes a breakdown of the ionic barrier of the muscle tissues, sending the signal to the transverse tubules. This process creates a

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MUAP, causing the muscle fibres to contract, following the sliding filament model [8], [9].

The muscle fibre action potentials summate spatially and temporally to form the MUAP. The continuous firing of the MUAP causes a continuous contraction, called the MUAP train (MUAPT). The cross-sectional area of the muscle fibre is 10-30 times smaller than the cross sectional area of the motor unit territory, and the motor units tend to overlap their fibre territories spatially [10], [11]; therefore, a single MUAPT (SMUAPT) can only be measured during very weak muscle contractions.

The MUAPT is the bio-electric signal which is recorded by an EMG device. The measured signal produced from a MUAP ranges from micro to milli-volts [12]. The amplitude of the MUAP is a function of muscle tissue, muscle fibre diameter, the distance the depolarisation must travel along the muscle fibre to the recording electrode, electrode properties and the subject's adipose tissue layer [12]. The adipose tissue layer acts as a low pass filter, attenuating high frequencies; therefore, the bio-electric signal has a lower measured amplitude with an increase in adipose tissue

III. The Benefits inherent with EMG

A. Advantages of EMG

Using the bio-electric signal detected through an EMG device for exoskeleton control can increase subject interaction, compared to standard kinematic and kinetic sensors alone [13], support a more natural unconscious interface compared to manual controllers (such as a joystick) [1], and the bio-electric signals are produced 2-3 ms before a muscle contraction occurs (this phenomenon is known as electromechanical delay) [12]: this allows for synchronous device and subject motion. The feasibility of this technology has been successfully demonstrated [14].

B. Limitations of EMG

The limitations associated with EMG must be well understood to produce a functioning device. One of the main limitations of EMG is its lack of long term stability due to electrical noise, electrode placement, subject fatigue and the electrode-skin interface. The two major forms of noise are due to EMI (the surrounding AC powered equipment is the dominating factor) and motion artefacts (the relative motion between the electrode and muscle tissue). Repetitive electrode placement is difficult due to limited physiological markers in close proximity to the desired recording site. Subject fatigue reduces the amplitude of the MUAP, resulting in a higher signal to noise ratio (SNR). The electrode-skin interface is variable due to perspiration, skin condition and hair layer depth [15]; therefore, the interface condition determines the electrode impedance, which shall be minimised. Crosstalk is the phenomenon of one MUAP's energy influencing the recording of another MUAP: this is due to the spatially overlapping fibre territories. This phenomenon makes it difficult to measure a single muscle site. The MUAPT recorded by an EMG device is not a direct measure of force, strength or effort. It is simply a measure of the electrical activity produced by the muscles as they contract; therefore, the recordings between each muscle group cannot be directly compared to one another. The differences in the recordings are due to the physical muscle size (length and width), the amount of muscle mass and the thickness of the adipose tissue beneath the recording electrode.

C. Alleviating the Limitations

Electrical noise can be reduced through the use of filters and a RLD circuit. The high impedance related with poor skin condition can be lessened with the removal of dead skin, cleaning the skin with alcohol and using electrode gel. Crosstalk can be reduced by using a double differential EMG design (via increased common mode rejection). To compare different muscle groups with one another, the MUAPT must be normalised: this normalisation is done using the subject's maximum voluntary contraction (MVC). An MVC is the maximum amount of energy a subject can voluntarily produce from a particular muscle group [15].

IV. Literature on Designing an EMG Device

Open source EMG circuitry is typically not discussed in literature. A small sample of publications outlining their design methods and considerations are discussed below; however, these publications have their limitations.

Naik and Zahak outline the bio-signal origin, various electrode types, general concerns associated with EMG recording and basic active filters [4]. The main concerns with this publication is that there is no full circuit diagram nor are there any results. A specific method for the removal of coupled EMI was also not mentioned: it appears that the design relies on the CMRR of the instrumentation amplifier to attenuate this interference, resulting in a reduced SNR.

Hardalaç and Canal outline their EMG design and the results they obtained through a trial containing 27 subjects [16]. The device had a RLD circuit which attenuated EMI; however, a notch filter was also implemented: thus defeating the inherent advantages of the RLD circuit (further discussion of the RLD is outlined in section VII). A complete circuit diagram was provided within the paper, but the design considerations behind the component choices were not clearly expressed: this hinders simple understanding of the circuit, prohibiting simple design modification.

Supuk et al. outlined their design and development approach for a low cost EMG device [17]. This contained their device considerations and results; however, the design has its limitations: the cost of the device was never stated and the main interference removal was done through a wavelet method. This approach is time consuming, 14 ms. As the electromechanical delay of human movement is 2-3 ms, the time associated with the signal processing prohibits the smooth, timely control of assistive exoskeleton devices. It was stated that their wavelet method had a better SNR than classical digital filters. From observation of their frequency spectrum analysis, it appears there was still a large peak at 50 Hz, which suggests there was still a large portion of AC mains EMI in the processed signal. This assumption could not be proven due to their diagram not having grid-lines, nor the frequency of the peak being outlined.

Youn and Kim developed a wireless EMG device [18]. Their paper contains a short section on circuit design and signal analysis. A complete circuit schematic was not provided. Once again the coupled EMI was attenuated via the use of software; however, this paper did not state the required computational time. Their results were compared to a commercial EMG device (DE 2-1, Delsys), and they reported that their device produced a higher SNR. The resulting frequency spectrum of both devices were shown, and even though they stated their device had a better SNR, the frequency analysis of the commercial device was far smoother. There were large peaks within the developed EMG's frequency analysis: this is once again assumed to be coupled EMI, but could not be proven from observation of their analysis.

The papers outlining EMG circuit design have their limitations: either the noise removal is inadequate due to lack of EMI attenuation; the computational processing time is too large; or the reasons behind design choices were never specified, resulting in a design which is difficult to understand and modify. Adjustable gain is not typically mentioned either, and this is rather important due to the different muscle groups and adipose layer thicknesses affecting the amplitude of the bio-signal. It is desirable to match the amplitude of the bio-signal to the data acquisition system to gain the highest possible resolution. De Luca [5], [19] outlines design considerations required to detect the MUAP, but a circuit diagram was never released.

V. System Overview

An EMG device was designed to produce a low cost method of retrieving the bio-electric signals of the skeletal muscles. The device can be broken down into four main stages,

1) Instrumentation Amplifier
2) Active Filtering
3) Right Leg Driver
4) Power and Isolation

The block diagram shown in Fig. 1 outlines a high level representation of a single channel from the EMG system. A single channel of the EMG device contains two inputs, and two outputs. The two inputs are the electrodes, 'Electrode +' and 'Electrode -', which measure the MUAP of a desired muscle group. One of the outputs, 'Reference Electrode', is used to attenuate induced EMI. The 'Output signal' is the signal of interest, the processed recording from the input electrodes. Only three main stages are explained in this paper: this is due to the assumption that the instrumentation amplifier is well understood.

VI. Hardware Filtering

The signal is processed to remove the redundant frequencies while being amplified. The filter applied to the signal is an active band-pass filter. The lower cutoff frequency is 20 Hz, which removes motion artefacts between the electrode and the patient's skin [12]. The upper cutoff frequency is 480 Hz, which is to remove unwanted signals produced by the tissue at the electrode site, without significant loss of motor unit potentials. The main issue with using a hardware filter with a fixed RC constant is that different muscle groups have different frequency ranges. The solution to this issue is to modify the upper cutoff frequency by having a variable RC constant: this can be achieved by using a variable resistor (not implemented in this design).

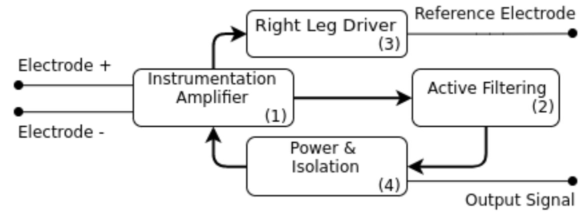

Fig. 1: A high level block diagram of a single channel of the EMG circuit outlined in this paper.

As the strength of the MUAP largely varies between subjects, the amplification gain of the band-pass filter was designed to be adjustable in the ranges of (4.3-104.3) $\frac{V}{V}$. The fixed gain of the instrumentation amplifier is 50 $\frac{V}{V}$; therefore, the system gain has a range of (215-5215) $\frac{V}{V}$.

A large amount of EMI gets coupled to the subject's body. The primary source of this EMI is the AC power supply, operating at 50 Hz (60 Hz USA). Some EMG designs incorporate a notch filter, with a bandstop of approximately 3 Hz, 1 Hz either side of the mains frequency (eg. 49-51 Hz). This technique has three main limitations. The first limitation is that the notch filter attenuates important information: the frequencies within close proximity to the AC mains frequencies. The second limitation is that the notch filter does not attenuate the multiple AC mains harmonics, reducing the SNR. The third limitation is that the AC mains frequency is country dependent; therefore, the hardware requires modification to suit the country of operation. A more suitable solution, which solves these issues, is the use a RLD circuit.

VII. Right Leg Driver (RLD)

The RLD circuit is designed to attenuate coupled EMI, not only AC mains frequency EMI, but also the associated harmonics. The circuit is commonly used with bio-potential differential amplifiers in an attempt to reduce common mode voltage. One of the largest issues encountered with common mode voltage is mismatched input impedance converting common mode voltage to differential mode voltage [20].

The RLD reduces EMI by applying the common mode voltage back into the subject's body, attenuating the EMI. This is achieved by inverting and amplifying the common mode voltage. This design uses the third electrode (the reference electrode) to provide a low impedance path between the subject and the amplifier common. Connecting the third electrode directly to common is undesirable: this can cause uncontrolled currents to flow through the subject.

The RLD contains three resistors which determines its performance characteristics: R_o, R_f and R_i. R_o is a current limiting resistor: limiting the maximum current that can flow from external sources though the subject. R_f is the feedback resistor, and when combined with R_i, defines the RLD gain (as the first op-amp is a voltage follower used to isolate the RLD from the instrumentation circuitry). The second op-amp is an inverting op-amp; therefore, the gain equation is defined:

$$A = \frac{V_o}{V_i} = \left(\frac{-R_f}{R_i}\right) \tag{1}$$

where V_o is the voltage measured at the reference electrode and V_i is the voltage at the inverting input of the op-amp.

A. RLD Amplification

The amplification factor of the RLD determines how well the circuit removes EMI. If this amplification factor is too low, there will be a noticeable amount of noise within the signal, reducing the SNR; however, if this factor is too large, the system becomes unstable, increasing the risk of saturation. The gain of the RLD is 40 $\frac{V}{V}$. This was chosen because it was a recommend design choice from [21]. The gain factor will ideally be tunable in the future, permitting increased adaptability to a larger range of measuring environments.

B. RLD Current Limiting

The R_o resistor is used to limit current flow through the driven right leg electrode, which is crucial when the circuit is not isolated from the mains voltage. If mains voltage appears on the subject, the current flow through the electrode leads must be less than 20 μA; therefore, if the circuit is not isolated, there must be atleast $12M\Omega$ of impedance between all of the amplifier leads and ground [20]. This protection resistance ensures safety of the subject in worst case scenarios; however, under normal operating conditions, the large impedance hinders the performance of the circuit. This is due to the common mode voltage being a function of the electrode's and limiting resistor's impedances; therefore, to reduce the common mode voltage, the impedance is required to be minimised. A method which permits a smaller impedance between all electrodes and ground, while maintaining a safe current limit, is the utilisation of an isolation amplifier.

VIII. Isolation

As subject safety is paramount, the EMG system is required to be an isolated system. This requires both an isolation amplifier and isolation of the power supply. Isolation amplifiers are utilised within medical instrumentation to permit subject and power supply leakage current isolation [22]. The isolation capacitance of these type of amplifiers can be used to limit the current flow through the subject,

$$C < \frac{I_{max}}{2\pi f V_{AC}} \tag{2}$$

where f is the frequency of the AC mains supply and I_{max} is the maximum current flow available on the electrode leads if mains voltage is apparent.

This means that the current limiting resistor is not required to limit current from external sources; however, the current limiting resistor is still typically included to limit transient currents that may flow on device power up. Typical values of R_o is within the range of 30 k Ω - 1 M Ω [20].

IX. Circuit Design

Using the aforementioned design considerations, a low cost ($30 unisolated, $75 isolated) prototype EMG device was designed, as shown in Fig. 2, and constructed. As the main portion of the MUAP is biphasic or triphasic [12], the device is required to have a dual power supply. This is provided by the isolated DC-DC converter.

Typically the data measured from an EMG device is recorded with the use of an oscilloscope or microprocessor attached to a computer (both of which are supplied via AC mains): these devices allow visual interpenetration of the recorded bio-potential signal. As bench top equipment is typically used, a DC power supply (which is also supplied via AC mains) is used to supply the EMG device. For these two reasons the isolation devices are required.

This solely hardware implemented design produces real time bio-potential signals produced by contracting skeletal muscles. If the signal is required to be an input for a control method of an exoskeleton device, post processing can be achieved through a microcontroller. Depending on the computation intensity of the control algorithm, this process can be achieved within the time of human electromechanical delay (2-3 ms), producing a smooth assistive device [1].

X. Data Collection and Analysis

This study was approved by the Human Ethics Committee, University of Canterbury (HEC 2015/53/LR-PS). An able-bodied subject's bicep brachii bio-potential signal was measured during an isometric contraction using the custom EMG device. Norotrode 20 bipolar surface EMG electrodes from Myotronics were used during the recording process. These electrodes are silver-silver chloride with silver snap connectors. The electrode contact surfaces come pre-gelled with water soluble hypoallergenic gel mounted on a hypoallergenic adhesive pad. Skin preparation was not performed before electrode placement.

A two part experiment was performed using a single subject to obtain data to demonstrate the performance of the EMG device and the benefit of the RLD. The subject was required to perform a series of isometric contractions of their right bicep brachii: a 0.8 second contraction followed by a two second pause. The experiment had two configurations which both involved modifying the reference electrode connection to the circuit: in the first configuration the reference electrode was connected to the RLD circuit, in the second configuration the reference electrode was connected to the circuit's ground (this is the common configuration for the reference electrode

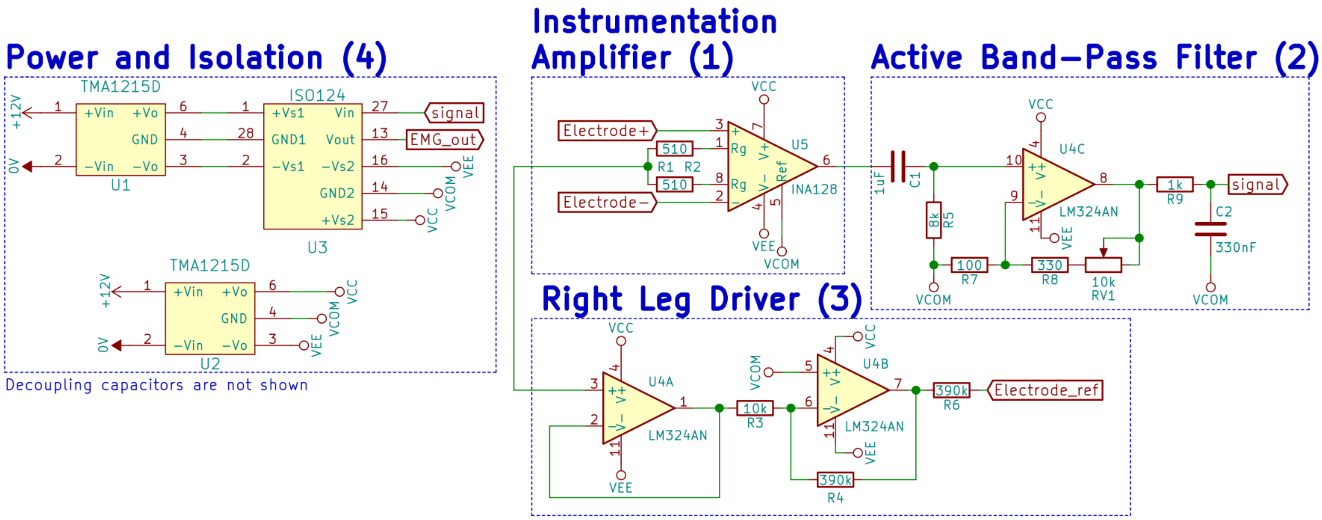

Fig. 2: A schematic showing a single channel of the prototype EMG device.

when the circuit does not utilise the addition of a RLD). The measuring of the two different configurations could not have been done at the same time: this is due to the RLD applying a current to the subject's body to attenuate the coupled EMI; however, the recording electrode pair position was not altered in any way during the recording process. During the experiment the strength and duration of the isometric contractions were attempted to be constant, and a rest period of two minutes was enforced between the two parts of the experiment: this was to minimise the effects of fatigue.

The data was used to plot four figures: raw EMG signal obtained with the addition of the RLD, Fig. 3, EMG power spectral density analysis (PSD) using the addition of the RLD Fig. 4, raw EMG signal obtained with the reference electrode connected to ground Fig. 5, EMG PSD with the reference electrode connected to ground Fig. 6. The data permits two methods of analysing the benefits inherent with the use of a RLD circuit. The raw signal data permits the calculation of the SNR for both reference electrode configurations, producing a numerical value to measure the benefit of the RLD. The frequency analysis is a visual representation of the attenuation ability of coupled EMI for both reference electrode configurations.

The isometric contractions were performed twelve times for each electrode configuration. The advantage of producing multiple contractions was the ability to select four of the most similar contractions between the two different configurations and take the mean of the four SNR.

From analysing the data, it was determined that the configuration utilising the RLD had a SNR of 57.8 dB and an amplitude of 241.9μV^2 at the AC mains frequency. The ground reference electrode configuration had a SNR of 63.7 dB and an amplitude of 1114μV^2 at the AC mains frequency. The data was recorded in a laboratory with grounded bench tops and floors, this reduces the amount of coupled AC mains EMI; therefore if this circuit was used in a electrically nosier environment, the SNR would be vastly reduced in the grounded configuration when compared to the RLD configuration. Comparing the two SNR, the values are very similar given the fact that the data cannot be taken from the same contraction; however, the values obtained from the PSD showed that the circuit utilising the RLD had a AC mains peak which was 78% less than the circuit using a ground reference configuration.

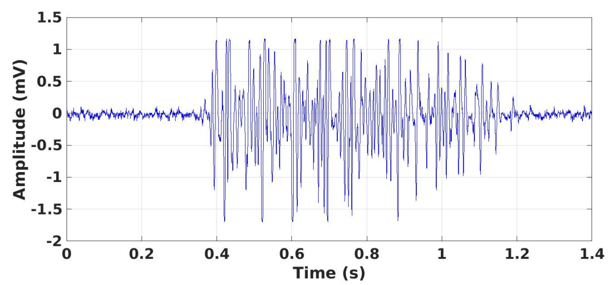

Fig. 3: EMG signal recorded during an isometric contraction of a subject's bicep brachii with the use of the EMG circuit prototype board, using the RLD reference configuration.

As this EMG device was designed due to the lack of access to a commercial EMG system, the measured data could not be directly compared to any commercially available technology. This does make it difficult to state the performance benefits of the system due to bio-potential recordings being very subject dependent. To the authors' knowledge there are no papers which outline the SNR obtained from the bicep brachii using a commercially available EMG system.

XI. Future Development

There are multiple functionalities which are still to be implemented: tunable RLD gain, shielded twisted pair electrode cables, decoupling capacitors and device compatibility. Shielded twisted pair cables reduce the effect of EMI coupling between the electrodes and the instrumentation amplifier. The

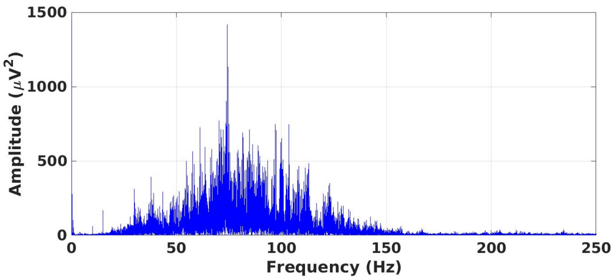

Fig. 4: Power spectral density analysis from twelve isometric contractions of a subject's bicep brachii with the EMG device configured to use the RLD reference electrode configuration.

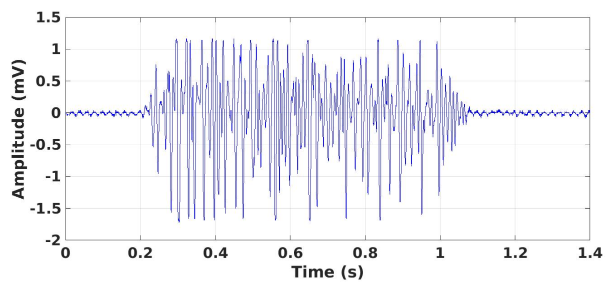

Fig. 5: A subject's bicep brachii muscle activity recorded during an isometric contraction using the EMG prototype board using the ground reference configuration.

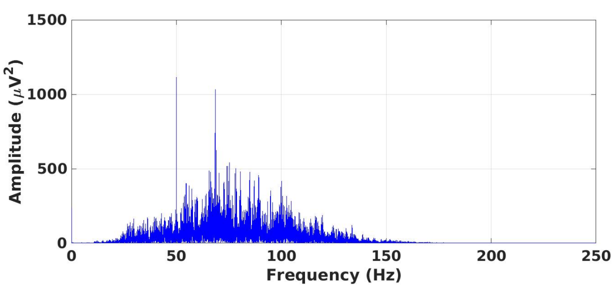

Fig. 6: Power spectral density analysis from twelve isometric contractions of a subject's bicep brachii with the EMG device configured to use the ground reference electrode.

current output signal ranges from $\pm 15V$. This device is to be used in conjunction with an FES device [23]; therefore, the output signal must be scaled to be compatible with the input range of the data acquisition system.

XII. Conclusion

The design and testing of a low-cost ($30 unisolated, $75 isolated) EMG device utilising a RLD driver circuit proved to be highly capable of recording bio-potential signals and attenuating coupled AC mains EMI. The circuit was used to record an isometric contraction of one subject's bicep brachii muscle using two different reference electrode configurations: ustilising the RLD and connected to the circuits ground. The SNR values were very similar bwteen the two contractions, 57.8 dB for the RLD reference configuration and 63.7 dB for the ground reference electrode configuration. A PSD was performed which outlined that the RLD attenuates coupled mains EMI, having a amplitude of $241.9\mu V^2$ which is 78% less than result obtained from the ground reference configuration.

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